HOUSE PRICE PREDICTION

A Machine Learning Based Web Application

Project Report

This report describes an educational web application that estimates house prices from selected property features and displays model evaluation results. The application uses Python, Pandas, NumPy, Scikit-learn, Streamlit and Plotly.

1. Introduction

House prices depend on several factors, including floor area, number of bedrooms and bathrooms, location, age, parking availability and floor number. This project demonstrates how regression algorithms can learn relationships between property features and a target price, then provide an estimated value through a simple web interface.

2. Objectives

• Create a user-friendly interface for entering property details.

• Train and compare multiple regression models.

• Show an estimated price and useful performance metrics.

• Visualize the data and model comparisons.

• Provide a clear, reproducible project that can be run locally.

3. Tools and Technologies

Technology	Purpose
Python	Main programming language
Pandas and NumPy	Data generation, organization and numerical operations
Scikit-learn	Preprocessing, model training and evaluation
Streamlit	Interactive web application interface
Plotly	Interactive data visualizations

4. Input Features

The application uses seven input features: area in square feet, bedrooms, bathrooms, location category (Urban, Suburban or Rural), house age in years, parking spaces and floor number. The target variable is house price expressed in lakh rupees.

5. Methodology

• Generate a reproducible synthetic dataset for demonstration.

• Separate the feature columns from the target price column.

• Split the dataset into training (80%) and test (20%) subsets.

• Impute missing values, standardize numeric columns and one-hot encode the location category.

• Train Linear Regression, Random Forest Regressor and Gradient Boosting Regressor models.

• Evaluate the models on the held-out test subset and select the model with the highest R^2 score for the dashboard prediction.

6. Model Evaluation

The dashboard reports three standard regression metrics. MAE (Mean Absolute Error) gives the average absolute difference between actual and predicted prices. RMSE (Root Mean Squared Error) gives extra weight to larger errors. R^2 indicates how much of the variation in prices is explained by the model. Lower MAE/RMSE and an R^2 closer to 1 are generally preferred. Exact values are calculated when the app runs.

7. Application Modules

• Home: enter house details and view the estimated price.

• Data Analysis: inspect the generated dataset, price distribution and average price by location; download the CSV.

• Model Performance: compare model metrics using tables and charts.

• About: review the objective, technologies, input features and limitations.

8. Limitations and Ethical Use

The bundled dataset is synthetic. It is generated for software demonstration and does not represent verified housing-market observations. The app's prices and performance metrics are therefore illustrative only. Do not use these estimates to make financial, legal or property-purchase decisions. A practical version requires a reliable, documented, legally obtained dataset from the relevant region, careful data cleaning, independent validation and periodic retraining.

9. Conclusion

This project demonstrates a complete beginner-friendly machine-learning workflow: data preparation, model training, evaluation, prediction and visualization in a web application. It can be extended with real housing data, additional features, model persistence and online deployment.

How to run: install dependencies using `pip install -r requirements.txt`, then execute `streamlit run app.py`.